

SHEILA MONERA CABARIQUE

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Work Authorization: EU Citizen (Spain) | UK Right to Work (Settled Status)

EDUCATION

Case Western Reserve University — Integrated B.S./M.S. Program Cleveland, OH
B.S. Computer Science & Data Science and Analytics, Secondary Major in Finance | GPA: 3.97 | Expected Dec 2027
Dean's High Honors all semesters, Tau Beta Pi Engineering Honor Society, NCAA Division III Varsity Athlete
Relevant Coursework: Algorithms, Machine Learning, Database Systems, Computer Security, Software Engineering, Computer Networks, Algorithmic Trading, Statistics

EXPERIENCE

European Spallation Source (ESS) Lund, Sweden
Research Software Intern May 2026 – Aug 2026

- Rebuilt a legacy analysis tool into a modular, vectorized architecture (NumPy/multiprocessing), improving processing speed 17.5x (7.0s to 0.4s)
- Integrated two new detector data pipelines onto the rewritten framework, extending its use across the group's instrumentation
- Redesigned the UI into a unified dashboard of all plots, replacing legacy Matplotlib pop-up windows

Research Software Intern May 2025 – Aug 2025

- Engineered a Python-based GUI for a detector data analysis tool, enhancing physicists' ability to test detectors and visualize data across multiple configurations
- Migrated Arduino sensor data transfer to a memory-efficient Telnet protocol and built a custom Python client to log and visualize sensor data; authored a comparative report on control protocols

TAPPAI Lab, Case Western Reserve University Cleveland, OH
Web Developer Mar 2026 – Present

- Architected and built the AI research lab's public website from scratch (Jekyll/Markdown) with a custom, scalable content structure
- Implemented a CI/CD pipeline via GitHub Pages for automated deployment; configured SEO and metadata to improve discoverability of the lab's research

Case Western Reserve University Cleveland, OH
Teaching Assistant — Financial Big Data Management Jan 2026 – May 2026

- Supported 50+ students via weekly office hours on Python-based data analysis, SQL queries, and financial modeling; graded assignments with detailed feedback

PROJECTS

StudySync — Academic Productivity Platform Spring 2026

- Project lead for a team of 4; architected a three-tier web app (React, FastAPI, PostgreSQL) deployed on a university HPC cluster, authoring a 16-page test plan for 100% requirement coverage
- Built a quorum-based scheduling algorithm integrated with the Google Calendar API (OAuth 2.0); implemented RBAC and real-time messaging via Firebase and WebSockets

University Event Management System Fall 2025

- UI/UX lead for a team of 5; built a React frontend with dynamic forms and real-time validation atop a Flask/PostgreSQL backend with 7 normalized tables and custom triggers for status automation
- Implemented end-to-end event browsing, registration, payment, and review workflows

LEADERSHIP & ACTIVITIES

CWRU Collaborative Coding Club Cleveland, OH

- Front End Project Lead (Jan 2024–Present) and former Treasurer (Sep 2024–Sep 2026) for a 300+ member club; led a front-end team while managing club finances and securing funding from the Undergraduate Student Government

NCAA Division III Varsity Swimming Cleveland, OH

- Balance 15+ hours/week of competitive training with a triple-major course load (3.97 GPA); qualified for the 2026 NCAA Division III Championships

Surf Life Saving (SLS) and Long Distance Surf Ski Poole, UK

- 20-time national medalist and qualified beach lifeguard; GB athlete selected for the 2022 World Championships (placed 14th); plan and lead group training sessions

SKILLS

Languages: Python, Java, SQL, C, R, JavaScript, HTML/CSS
Tools & Frameworks: React, FastAPI, PostgreSQL, Firebase, NumPy/Pandas, Git
Certificates: Forté Career Ready Certificate Program, Expedition EY Certificate Program, RLSS Lifeguard Certification
Spoken Languages: English & Spanish (native), French (B2)